KARTHIK VEERLA

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PROFESSIONAL SUMMARY

- Full Stack Software Engineer with around **5 years of experience** designing, developing, and deploying **scalable enterprise web applications** using **Java, Spring Boot, Hibernate, React.js, Python/Django, and Node.js/TypeScript**.
- Solid knowledge of **OOP, data structures & algorithms**, **exception handling, generics, streams, lambdas**, and the Java **Collections** Framework to build maintainable and high-performance systems.
- Designed and developed **RESTful APIs and microservices** using **Spring Boot, Spring MVC, Spring Data JPA, Hibernate, Spring Security and Spring Cloud**, applying **dependency injection, exception handling**, and modular architecture principles.
- Modernized **legacy Java systems (Servlets, JSP, XML)** into scalable **microservices** architectures, improving performance, maintainability, and deployment efficiency.
- Engineered **parallel processing pipelines** using asynchronous programming (CompletableFuture, Async/Await) to perform **multi-URL accessibility scans**, reducing execution time by over 60%.
- Proficient in **Python and Django** for building backend services, automation scripts, and rapid API/admin interfaces integrated with distributed systems.
- Skilled in **frontend development** using **React.js, Redux, Context API, Hooks, TypeScript, JavaScript, HTML5, CSS3**, and UI frameworks like **Bootstrap and Tailwind CSS** for dynamic and accessible user interfaces.
- Integrated **OpenAI** and **Claude** APIs to enable **AI-driven** analysis, automation, and intelligent validation across application workflows, enhancing system efficiency and decision-making accuracy.
- Proficient in **API communication & security** using RESTful integrations, JSON handling, JWT/OAuth2 authentication, and Spring Security custom filters.
- Worked with XML for project configurations, data serialization, and API integrations, including Spring and legacy system setups requiring **XML-based mappings** and schema validations.
- Experienced in **databases** including **Oracle, MySQL, PostgreSQL, MongoDB, DynamoDB, and Redis**, leveraging **JPA/Hibernate, JPQL**, and native SQL for efficient data operations.
- Implemented **CI/CD pipelines** using **Docker, Kubernetes, Jenkins, and GitHub Actions**, deploying on **AWS (EC2, S3, RDS, DynamoDB, Lambda, CloudWatch)** and enabling automated builds and monitoring.
- Practiced **Agile/Scrum** with **Jira, Git Version Control**, delivering secure, accessible, and production-grade solutions with **unit/integration testing (JUnit, Mockito, Selenium, Playwright)** and **SOLID design principles**.

EDUCATION

MASTER'S DEGREE | Computer and Information Sciences | University at Albany (SUNY)
BACHELOR'S DEGREE | Computer Science and Engineering | KLUniversity, Vijayawada

TECHNICAL SKILLS

Programming & Scripting Languages: Java, Python, JavaScript, TypeScript, C#, C, HTML5, CSS3, SQL.

Frameworks & Platforms: Spring Boot, Spring MVC, Spring Data JPA, Hibernate, Spring Security, Django, React.js, Node.js, Express.js, Kafka, Redis.

Databases & ORM: Oracle SQL, MySQL, PostgreSQL, SQL Server, MongoDB, DynamoDB, Redis, Hibernate/JPA.
Testing & Automation: JUnit, Mockito, Playwright, Selenium, Postman, axe-core, WCAG 2.1 AA Compliance
Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, CloudWatch, SNS, SQS), Docker, Kubernetes, Jenkins, Git, GitHub.
Tools & Utilities: Eclipse IDE, Visual Studio Code, Maven, npm, Jira, Postman, XML (configurations, data serialization)
Software Development Practices: OOP, SOLID Principles, TDD, CI/CD Pipelines, RESTful API Design, Microservices Architecture, Agile (Scrum/Kanban), Version Control, Accessibility Engineering.
AI Integration & Intelligent Systems: OpenAI API, Claude API, AI-driven Validation Pipelines, Automated Audit Reporting, DOM–UI Analysis.

EXPERIENCE

SOFTWARE DEVELOPER | Xove Consulting Services | Raleigh, NC **May 2025 – current**

- Engineered a scalable accessibility audit platform using **Java 17, Spring Boot, and Spring MVC**, enabling automated **WCAG 2.1 AA compliance validation** across enterprise web pages by crawling user-submitted domains, scanning internal URLs, and structuring results as JSON for downstream analytics.
- Designed and orchestrated **asynchronous scan pipelines** using Spring Async and CompletableFuture, achieving **parallel audits** across multiple pages and reducing total execution time by **60%** in benchmarked environments.
- Built **dynamic React.js dashboards** with Tailwind CSS, visualizing scan summaries, violation trends, and page-level accessibility scores, empowering QA and compliance teams with actionable insights.
- Implemented **intelligent URL crawling** using jsoup and domain filters to recursively discover internal links, feeding them into the **axe-core audit engine** via Selenium WebDriver for automated rule-based validation.
- Integrated **axe-selenium-java** for DOM-level accessibility scanning, injecting axe-core into browser sessions and capturing violations with structured metadata including severity, rule ID, and impacted elements.
- Persisted scan results, audit history, and metadata in **MongoDB** optimizing for **fast aggregation** and compliance reporting across large-scale multi-page audits.
- Developed **Playwright-based automation** in TypeScript for advanced accessibility validation, covering **keyboard navigation, ARIA roles, semantic structure**, and dynamic content rendering across SPAs.
- Leveraged **OpenAI and Claude APIs** for **AI-driven accessibility heuristics**, detecting **DOM–UI mismatches** and edge-case violations beyond rule-based scanning to enhance coverage and reduce false negatives.
- Worked with **XML-based configuration files** for crawler rules, scan profiles, and engine tuning, ensuring flexible setup and environment-specific execution.

SOFTWARE DEVELOPMENT SPECIALIST | New York State Dept of Health | Albany, NY **Oct 2024 – May 2025**

- Developed an internal data management system for laboratory experiment results, replacing manual Excel workflows with automated, secure, and queryable web interfaces.
- Built scalable full-stack modules using **Python, Django, React.js, and Oracle SQL**, improving data processing efficiency by **32%** through structured pipelines and optimized database operations.
- **Integrated third-party RESTful APIs and web services for laboratory reference data**, automating data synchronization and reducing manual data reconciliation time by **28%**.
- Engineered robust data ingestion and transformation pipelines to collect experiment data from lab scientists via web forms and CSV uploads, applying Python-based cleaning, mapping, and validation scripts.
- Designed and implemented a **modular Django backend** integrated with **AWS Lambda, RDS, and S3**, ensuring scalability, secure storage, and high data throughput for ongoing experiments.
- Developed **responsive React.js front-end components** with dynamic filtering, search, and analytics dashboards for tracking experiments by type, date, and biological metrics.
- Deployed containerized applications using **Docker** and **Jenkins CI/CD pipelines**, ensuring consistent build automation and seamless cloud deployment.
- Collaborated in an **Agile/Scrum** setup, writing **unit and integration tests (PyTest, Postman)** and enforcing **SOLID principles** for maintainable, modular system design.

SOFTWARE DEVELOPER | New York State Office of ITS | Albany, NY **Jan 2024 – Sep 2024**

- Contributed to the modernization of the **unified NY.gov home portal**, enabling seamless access to multiple New York State agency services by redesigning core authentication workflows and improving system scalability.
- Refactored legacy Java modules (Servlets, JSP, XML) into **Spring Boot–based microservices** using **Hibernate** and **REST APIs**, reducing response time by **35%** and enhancing cross-agency data processing efficiency.
- Enhanced authentication and session management for multi-agency logins, integrating **Spring Security**, **JWT**, and **SAML-based Single Sign-On (SSO)** to support secure, centralized user access across 300+ applications.
- Optimized **SQL queries and caching strategies**, applying indexing, connection pooling, and query refinements to reduce backend latency and improve overall database throughput.

- Utilized **Apache Maven** for build automation and dependency management; maintained development workflows in **Eclipse** with **Git** and **Azure DevOps** for version control and continuous delivery.
- Deployed scalable cloud solutions using **AWS ECS, Lambda, and S3**, strengthening system resilience and availability for **35M+ NY.gov user accounts**.

RESEARCH ASSISTANT | SUNY Research Foundation | Albany, NY **Jan 2023 – Dec 2023**

- Contributed to the development of a **Research Data Management Portal** built using **Python (Django)** and **ReactJS**, supporting faculty and student researchers in securely uploading, processing, and visualizing experimental datasets.
- Collaborated with a team of developers and researchers to implement **RESTful APIs** for handling experiment metadata, analytical results, and visualization endpoints, improving research workflow efficiency by **31%**.
- Assisted in integrating **third-party APIs and open research datasets** (e.g., PubMed, NIH) for automated literature correlation and enrichment of research data, reducing manual effort by **24%**.
- Supported automation of **data processing pipelines** using **Python scripts and AWS Lambda**, increasing task execution speed and scalability while maintaining accuracy for recurring data updates.
- Helped deploy Django applications on **AWS Lambda** with **API Gateway**, optimizing performance and minimizing infrastructure costs through a serverless setup.
- Developed and maintained **React.js components** and **UI dashboards** for research data visualization using **Chart.js** and **Tailwind CSS**, improving interactivity and responsiveness.
- Participated in **frontend testing** using **Jest** and **React Testing Library**, ensuring stable UI performance and consistent component behavior across updates.
- Worked on implementing **JWT-based authentication** and **role-based access control** within the Django REST Framework to ensure secure access for faculty and research staff.
- Contributed to team sprints under **Agile methodology**, managed progress using **Jira**, and collaborated via **GitHub** for version control and code reviews.

SOFTWARE DEVELOPER | Hexagon Capability Center India | Hyderabad, INDIA **Aug 2020 – Aug 2022**

- Developed and optimized **microservices-based architecture** using **Java, Spring Boot, and Hibernate**, enhancing data synchronization and processing efficiency by **27%** across distributed enterprise systems.
- **Diagnosed and resolved complex frontend and backend issues**, achieving a **93% bug resolution rate** and reducing customer support tickets by **24%** through proactive debugging and performance tuning.
- Redesigned and implemented **React-based dialog components** across multiple application modules, significantly improving **UI consistency, responsiveness, and user experience**.
- Integrated **interactive maps** into the OnCallDispatch platform using **Map APIs**, enabling **real-time incident visualization and sketching tools** for field operations and dispatch workflows.
- Utilized **Hibernate and Spring Data JPA** for ORM, streamlining database access and minimizing boilerplate code through reusable repository patterns.
- Leveraged **XML configurations** for data exchange and service integration across legacy and modernized components within the Hexagon OnCallDispatch ecosystem.
- Performed in-depth **code reviews, refactoring, and backend optimization** in Java and Spring Boot, reducing **REST API response time by 28%** and improving scalability.
- Architected and deployed **cloud-native applications** using **Docker and Kubernetes**, optimizing resource allocation and improving deployment reliability across multiple environments.
- Implemented **CI/CD pipelines** with **Azure DevOps**, ensuring seamless automated builds, testing, and deployments with zero manual intervention.
- Enhanced software quality through **unit and integration testing** using **JUnit and Mockito**, increasing code coverage by **35%** and reducing post-release defects.
- Collaborated closely with **cross-functional teams** using **Azure DevOps, Visual Studio, and Git**, ensuring **on-time deliverables** and maintaining 99.9% alignment with evolving business requirements.